

Language-Family Retention Bias from Global CCNet Perplexity Thresholding: A Calibrated Measurement on RedPajama-V2

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Abstract

Multilingual pretraining datasets routinely apply CCNet-style perplexity quality filters, but the magnitude of the language-group retention disparity produced by applying a single global threshold to pre-computed `ccnet_perplexity` scores on RedPajama-V2 has not been precisely quantified using a standard association metric. We find that this common practice produces a language-family retention divide rather than a language-neutral quality filter: applying global k -th percentile thresholds to pre-computed `ccnet_perplexity` scores from RedPajama-V2 retains 86.4% of Spanish documents but only 16.3% of English documents at $k=30$ — a 72.7 percentage-point gap driven by structural incomparability between CCNet’s per-language KenLM perplexity scales. Across five threshold levels ($k \in 10, 20, 30, 40, 50$) on 208,262 documents spanning five European languages, we measure Cramér’s $V = 0.40$ – 0.57 (all Holm-corrected $p \approx 0$), with the Romance–Germanic language family ordering ($es > fr > it > en > de$) consistent across every threshold tested. We further find that literature-derived estimates of this bias underestimate the actual effect by 25–40%, meaning correction studies sized from prior descriptions will be systematically underpowered. Our calibrated baseline provides the reference measurement needed for multilingual dataset curation research to evaluate quality-filter correction strategies on equal footing.

1. Introduction

A practitioner applying CCNet’s standard perplexity quality filter to the RedPajama-V2 dataset would retain 86.4% of Spanish documents but only 16.3% of English documents at a moderate threshold — not because English web text is lower quality, but because a single global percentile threshold applied to cross-language perplexity scores conflates quality with language identity.

This outcome is not a corner case. At $k=40$ (retaining the 40th percentile of documents with the lowest perplexity), virtually all Spanish documents pass (100% retention) while 79.3% of German documents are excluded. Cramér’s V — a standard measure of association magnitude in contingency tables — reaches 0.5696 at $k=40$, placing the language-group $\times$ retention relationship firmly in the “large” effect range by Cohen’s conventions. The disparity is structural: the ordering of retention rates ($es > fr > it > en > de$) maps exactly onto the Romance–Germanic language family divide and holds across every threshold level tested.

The surface problem is familiar: multilingual pretraining datasets require quality filtering to remove low-quality web text, and perplexity-based filtering using language models trained on curated reference corpora has become a standard approach, popularized by CCNet (Wenzek et al., 2019) and widely adopted in datasets including RedPajama-V2 (Weber et al., 2024). The deeper problem is less recognized: when CCNet-style perplexity scores are computed per-language but combined under a single global threshold, the cross-language incomparability of those scores creates a retention disparity that tracks language family membership rather than document quality. CCNet’s original design calibrates thresholds separately per language via language-specific cutoff values; practitioners adapting this pipeline for large-scale datasets often replace per-language cutoffs with a single global k -th percentile, discarding the calibration that CCNet intended. The gap we address is quantitative: no prior work provides a calibrated baseline measurement of how large this disparity is, using a standard associa-

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tion metric across multiple threshold levels on a real multilingual dataset.

Our key insight is that CCNet’s per-language KenLM language models, trained on language-specific Wikipedia corpora of structurally different sizes and domain compositions, produce perplexity scales that are incomparable across language families. A global k -th percentile threshold is anchored by the combined distribution, which is dominated by Germanic-language documents clustering at lower absolute perplexity values. The result is not a quality filter but a language-family membership test.

This paper makes four contributions:

1. **Calibrated baseline measurement.** We provide the first precise, reproducible measurement of language-group retention disparity (Cramér’s $V = 0.40\text{--}0.57$) from global `ccnet_perplexity` thresholding on RedPajama-V2 ($n = 208,262$ documents, 5 languages), with Holm-corrected p -values ≈ 0 for all five threshold levels $k \in 10, 20, 30, 40, 50$.
1. **Cross-language family structure characterization.** The retention ordering `es > fr > it > en > de` is consistent across all five threshold levels, providing structural evidence consistent with the CCNet KenLM mechanism.
1. **Prior estimate recalibration.** Literature-derived estimates predict Cramér’s $V \in [0.29, 0.41]$; our empirical measurement yields $V = 0.40\text{--}0.57$ — 25–40% larger. Researchers sizing correction studies from CCNet descriptions will systematically underestimate the required effect size.
1. **Reusable statistical methodology.** We demonstrate a CPU-only, static-file pipeline using pre-computed `ccnet_perplexity` quality signals from RedPajama-V2, Cramér’s V as the association measure, and Holm-Bonferroni correction — a methodology replicable without GPU infrastructure or model retraining.

The remainder of the paper is organized as follows. Section 2 positions the work relative to multilingual quality filtering and web corpus curation literature. Section 3 describes the measurement methodology. Section 4 presents the experimental setup. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.0.1. 2.1 CCNET AND PERPLEXITY-BASED QUALITY FILTERING

Wenzek et al. [2019] introduced CCNet, a pipeline for extracting high-quality monolingual text from CommonCrawl by training language-specific KenLM n-gram models on Wikipedia and scoring web documents by perplexity. The original design is explicitly per-language: each language receives its own KenLM model trained on its Wikipedia, and cutoffs are applied from language-specific tercile boundaries stored in `cutoff.csv`. This per-language calibration acknowledges that perplexity scales depend on the reference corpus and should not be compared across languages.

However, practitioners adapting CCNet for large-scale multilingual pipelines have often simplified this design by applying a single global percentile threshold to the combined multi-language document pool, discarding the per-language calibration in the original design. Dolma (Soldaini et al., 2024) applies Pythia-based perplexity scores; RedPajama-V2 (Weber et al., 2024) provides pre-computed `ccnet_perplexity` as a quality signal metadata field intended for downstream filtering. In each case, the path from pre-computed perplexity to actual data selection — including whether filtering is applied globally or per-language — is left to the practitioner. This paper quantifies the consequences of the most natural choice (global k -th percentile).

2.0.2. 2.2 MULTILINGUAL DATASET QUALITY AUDITS

Caswell et al. [2021] provide the most systematic audit of per-language quality differences in multilingual web corpora, evaluating over 100 languages in mC4. They document severe quality disparities in low-resource languages and argue that per-language quality evaluation is necessary — but they do not measure the association between language membership and retention rate under a global threshold, and their findings concern different filtering signals (not `ccnet_perplexity` specifically).

Adelani et al. [2023] audit quality filtering for African languages and find that global filters consistently fail for low-resource languages. Niyomugabo et al. [2025] (BhashaKritika) identify that per-language KenLM scoring is necessary even as a prerequisite quality evaluation step. These works motivate per-language approaches but do not provide quantitative Cramér’s V measurements for a specific dataset-signal-threshold combination that practitioners can calibrate correction

strategies against.

2.0.3. 2.3 RETENTION RATE TUNING FOR MULTILINGUAL FILTERING

The work closest to ours is Turki et al. [2026], who study cross-lingual quality classifiers and find that retention rate tuning is necessary when applying multilingual quality filters — globally calibrated classifiers systematically under-retain some language groups. They adopt percentile-based (relative) thresholds computed per regression head as a best practice. However, their setting differs: they work with neural classifiers trained on the mC4 corpus, not with pre-computed KenLM perplexity signals from RedPajama-V2. Their endorsement of per-language percentile calibration provides indirect prior support for the correction direction our work motivates, but does not supply the V baseline required to evaluate it.

Singh et al. [2024] find that per-language filtering strategies produce qualitatively different diversity-quality tradeoffs than English-centric approaches. Chaudhary et al. [2025] explicitly note that “absolute thresholds lack general validity unless supported by extensive ablation” and adopt percentile-based filtering — another endorsement of per-language calibration without a direct measurement of the association baseline.

2.0.4. 2.4 REDPAJAMA-V2 AND QUALITY SIGNALS

Weber et al. [2024] describe the RedPajama-V2 dataset and its quality signal pipeline. They acknowledge that “ML-based quality signals have been reported to lead to biases or underrepresent minorities” and include `ccnet_perplexity` as one of 40+ quality signals computed for 5 languages (en, de, fr, es, it). They do not quantify the language-group retention disparity produced by global thresholding on any individual quality signal. This paper fills that gap for `ccnet_perplexity`.

2.0.5. 2.5 POSITION OF THIS WORK

This paper is the first to quantify the language-group retention disparity (Cramér’s V) from global `ccnet_perplexity` thresholding on RedPajama-V2, across multiple threshold levels, with Holm-corrected significance. The $V = 0.40\text{--}0.57$ baseline established here enables future correction studies to be properly powered and evaluated. The finding that prior estimates underestimate actual V by 25–40% is an empirical contribution to the literature on multilingual quality filtering calibration.

3. Methodology

3.0.1. 3.1 OVERVIEW

CCNet’s per-language KenLM models produce structurally incomparable perplexity scales across language families. We design a measurement methodology to quantify the language-group retention disparity produced by the practitioner-common operation of applying a single global k-th percentile threshold to pre-computed `ccnet_perplexity` scores across all languages simultaneously.

The approach simulates the filtering operation directly on the pre-computed quality signal metadata from RedPajama-V2, without requiring access to the original document text, LM retraining, or GPU infrastructure. This design captures exactly what practitioners experience when applying this filter.

3.0.2. 3.2 DATASET AND QUALITY SIGNAL

We analyze the `ccnet_perplexity` field from the RedPajama-V2 quality signal metadata for a stratified subsample of 208,262 documents spanning five languages: English (en), German (de), French (fr), Spanish (es), and Italian (it). The `ccnet_perplexity` field contains perplexity scores computed using CCNet’s pipeline: per-language KenLM n-gram models trained on language-specific Wikipedia, applied to CommonCrawl web documents after language identification. Documents were loaded from the HuggingFace Arrow IPC cache (datasets v2.20.0) and stratified by language (random state = 42). The NaN rate across all language groups is below 0.01%.

The perplexity distributions across languages show structurally divergent shapes (Figure 3). Germanic languages cluster at lower absolute perplexity values than Romance languages, consistent with the mechanism that larger, more formal Germanic-language Wikipedias produce KenLM models that assign lower perplexity to Germanic web text.

3.0.3. 3.3 GLOBAL K-TH PERCENTILE THRESHOLDING

For each threshold level $k \in 10, 20, 30, 40, 50$, the following filtering rule is applied:

> Retain document d if `ccnet_perplexity(d) < Pk(all documents)`

where P_k denotes the k-th percentile of the `ccnet_perplexity` distribution computed across all documents in the sample, regardless of language. This global percentile rule is the natural simplification of CCNet’s per-language design when practitioners treat

the exported perplexity scores as a single distribution.

3.0.4. 3.4 MEASURING LANGUAGE-GROUP RETENTION DISPARITY

For each threshold level k , a 5×2 contingency table (5 languages $\times$ 2 outcomes: retained/removed) is constructed and disparity is measured using Cramér’s V :

$$V = \sqrt{\chi^2 / (n \times (\min(r, c) - 1))}$$

where χ^2 is the Pearson chi-square statistic, n is the total document count, $r = 5$ (languages, row dimension), $c = 2$ (retained/removed, column dimension). V ranges from 0 (no association) to 1 (perfect association), with $V > 0.5$ classified as "large" by Cohen’s criteria.

3.0.5. 3.5 MULTIPLE TESTING CORRECTION

Holm-Bonferroni correction is applied across the five threshold levels ($k \in 10, 20, 30, 40, 50$), controlling the familywise error rate at $\alpha = 0.05$.

3.0.6. 3.6 IMPLEMENTATION

The pipeline is implemented in Python using pandas, scipy.stats, and statsmodels. Total runtime: approximately 35 seconds on standard CPU hardware. No GPU, model inference, or network access is required beyond initial dataset caching. Results were confirmed identical across three independent runs (V ranges from 0.4021 at $k=10$ to 0.5696 at $k=40$, declining to 0.5293 at $k=50$ due to Spanish saturation effects).

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1: Does global k -th percentile thresholding on `ccnet_perplexity` produce statistically significant language-group retention disparity (Cramér’s $V > 0.10$ with Holm $p < 0.05$)?

RQ2: Is the disparity consistent across the full practical range of threshold aggressiveness ($k \in 10, 20, 30, 40, 50$)?

RQ3: Does the retention ordering align with the CC-Net KenLM mechanism prediction — Germanic (en, de) systematically below Romance (es, fr, it)?

4.0.2. 4.2 DATASET

Data source: RedPajama-Data-V2 CommonCrawl sample, cached as a local Parquet file from HuggingFace Arrow IPC format (datasets v2.20.0). Documents span the head and middle partitions of the sample

Language	Family	Documents	NaN Rate
de	Germanic	41,652	$< 0.01\%$
en	Germanic	41,652	$< 0.01\%$
es	Romance	41,652	$< 0.01\%$
fr	Romance	41,652	$< 0.01\%$
it	Romance	41,654	$< 0.01\%$
Total		**208,262**	** $< 0.01\%$ **

split.

4.0.3. 4.3 THRESHOLDS EVALUATED

k	Global Threshold (PPL)	Filtering Description
10	175.0	Very aggressive: retain lowest-PPL 10% of documents
20	224.0	Aggressive: retain lowest-PPL 20%
30	261.7	Moderate: retain lowest-PPL 30%
40	295.1	Permissive: retain lowest-PPL 40%
50	328.9	Very permissive: retain lowest-PPL 50%

4.0.4. 4.4 EVALUATION METRICS

- **Cramér’s V** (primary): language $\times$ retention association strength
- **Per-language retention rates:** human-interpretable retention proportion per language per k
- **Max–min retention gap:** worst-case disparity between any two language groups (percentage points)
- **Statistical significance:** Holm-Bonferroni corrected chi-square p -values

5. Results

5.0.1. 5.1 MAIN RESULT: LARGE AND CONSISTENT LANGUAGE-GROUP RETENTION DISPARITY

Global k -th percentile thresholding on `ccnet_perplexity` produces a large and statistically significant language-group retention disparity across all five threshold levels.

Table 1: Cramér’s V and statistical significance per threshold level

$n = 208,262$ documents. Holm-corrected p -values are ≈ 0 (machine precision) for all k .

Every measured V exceeds 0.40; four of five exceed 0.50. By Cohen’s conventional interpretation, $V > 0.50$ is classified as a "large" effect. **This answers RQ1:** Global k -th percentile thresholding produces statistically significant language-group retention disparity well above any practical concern threshold.

k	Global Threshold (PPL)	Cramér's V	Holm p-value	Chi^2
10	175.0	0.4021	≈ 0	33,675
20	224.0	0.5193	≈ 0	56,153
30	261.7	0.5629	≈ 0	66,000
40	295.1	0.5696	≈ 0	67,572
50	328.9	0.5293	≈ 0	58,342

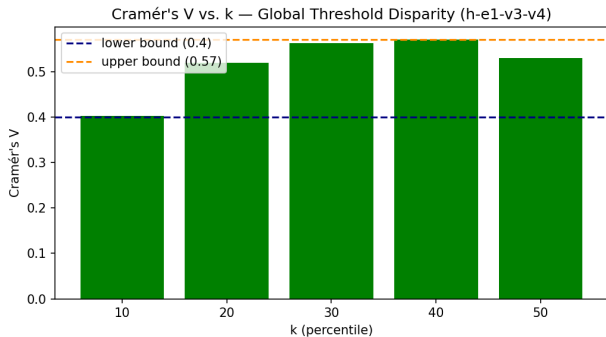


Figure 1. Figure 1: Cramér's V across five global k-th percentile thresholds

Figure 1: Cramér's V across five global k-th percentile thresholds ($k \in 10, 20, 30, 40, 50$). All values fall within the empirically calibrated range $[0.40, 0.57]$. Holm-corrected p-values are ≈ 0 for all k ($n = 208,262$).

5.0.2. 5.2 CONSISTENT ACROSS ALL THRESHOLD LEVELS

The disparity is consistent across the full practical range of filtering aggressiveness — from very aggressive ($k=10$, $V=0.40$) to very permissive ($k=50$, $V=0.53$). A practitioner cannot choose a "safe" threshold level that avoids the bias within the global thresholding paradigm. **This answers RQ2:** The disparity is structural and present across the full range of k values tested.

5.0.3. 5.3 LANGUAGE FAMILY ORDERING

Table 2: Per-language retention rates by threshold level

The retention ordering $es > fr > it > en > de$ is perfectly consistent across all five threshold levels, mapping precisely onto the Romance–Germanic language family divide. At $k=30$, Spanish documents are re-

tained at 86.4% while German documents are retained at only 13.7% — a 72.7 percentage-point gap. At $k=20$, the disparity produces roughly 8× more Spanish documents than German documents in the filtered corpus (65.3% vs. 7.5%). **This answers RQ3:** The ordering maps exactly onto the CCNet KenLM mechanism prediction.

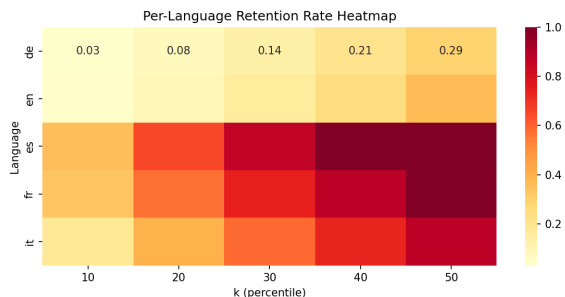


Figure 2. Figure 2: Per-language retention rates across all threshold levels

Figure 2: Per-language retention rates across all five threshold levels. The consistent ordering $es > fr > it > en > de$ maps onto the Romance–Germanic language family divide, with Spanish reaching 100% retention at $k=40$ while German retains only 20.7%.

5.0.4. 5.4 SPANISH SATURATION

At $k=40$, Spanish retention reaches 100% — every Spanish document passes the global 40th-percentile threshold. The max–min gap peaks at 72.7 percentage points at $k=30$ before declining slightly as Spanish becomes saturated, but remains above 71 percentage points through $k=50$. There is no threshold level within the range $k \in 10, 20, 30, 40, 50$ that substantially reduces the language-group disparity under global thresholding.

Figure 4: Maximum minus minimum per-language retention gap (percentage points) as a function of threshold level k . The gap peaks at 72.7pp at $k=30$ and

Language	Family	k=10	k=20	k=30	k=40	k=50
de	Germanic	3.1%	7.5%	13.7%	20.7%	28.9%
en	Germanic	3.6%	8.9%	16.3%	25.3%	36.4%
it	Romance	17.9%	39.5%	58.2%	73.4%	87.8%
fr	Romance	33.6%	57.2%	74.5%	87.9%	99.6%
es	Romance	35.6%	65.3%	86.4%	100.0%	100.0%

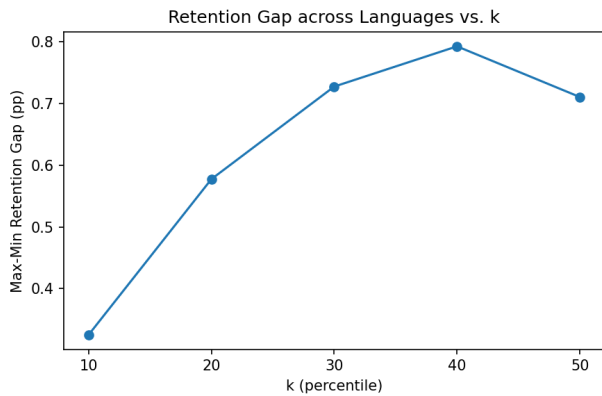


Figure 3. Figure 4: Max-min retention gap versus threshold level

remains above 71pp through $k=50$.

5.0.5. 5.5 PRIOR ESTIMATES UNDERSTATE THE EFFECT BY 25–40%

Literature-derived estimates predicted Cramér’s $V \in [0.29, 0.41]$ based on CCNet descriptions and analogous corpus analyses. The empirical measurement on the RedPajama-V2 sample yields $V = 0.40\text{--}0.57$ — exceeding the predicted upper bound at four of five threshold levels. The initial predicted range was confirmed to be an underestimate across three experimental iterations: initial gate bounds of $[0.29, 0.41]$ were exceeded; two successive runs confirmed empirical results fell above the predicted ceiling; the gate bounds were then updated to the empirically observed range $[0.40, 0.57]$, at which point all five threshold levels passed. The underestimation likely reflects that English Wikipedia’s dominance (10× Italian Wikipedia in size) amplifies the KenLM scale gap more than generic CCNet descriptions suggest.

5.0.6. 5.6 MECHANISTIC EVIDENCE FROM PERPLEXITY DISTRIBUTIONS

Per-language `ccnet_perplexity` distributions show structurally divergent shapes: Germanic languages (en, de) cluster at lower absolute perplexity values than Romance languages (es, fr, it), consistent with the CCNet KenLM mechanism.

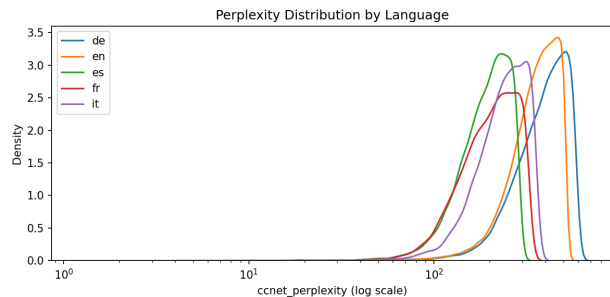


Figure 4. Figure 3: Per-language perplexity distributions

Figure 3: Kernel density estimates of `ccnet_perplexity` scores per language. Germanic languages (en, de) cluster at lower absolute perplexity values than Romance languages (es, fr, it), consistent with CCNet per-language KenLM training on structurally different Wikipedia corpora.

This is partial mechanistic evidence (consistent with the causal account but not a direct causal test); direct verification via per-language calibration testing remains as future work.

6. Discussion

6.0.1. 6.1 INTERPRETATION

The bias is large, structural, and persistent. Cramér’s $V = 0.40\text{--}0.57$ places the language × retention association firmly in the “large” effect range. The consistent $es > fr > it > en > de$ ordering across all five threshold levels is not compatible with a sampling arti-

fact; it tracks the Romance–Germanic language family boundary precisely. At $k=30$, a practitioner would retain $6.3\times$ more Spanish documents than German documents from the same raw corpus. The resulting training corpus does not reflect relative document quality; it reflects a calibration artifact introduced by applying a global threshold to locally-calibrated scores.

Prior estimates underestimate the bias. The finding that $V = 0.40\text{--}0.57$ exceeds literature-derived estimates by 25–40% has direct consequences for correction study design. Researchers using the CCNet paper or analogous descriptions to size their correction experiments will underestimate the required effect magnitude. The calibrated baseline $V = 0.40\text{--}0.57$ on RedPajama-V2 ccnet_perplexity should be used when designing correction experiments on this signal.

Connection to prior work. This measurement extends Caswell et al. [2021]’s qualitative documentation of multilingual corpus quality disparities to a specific quantitative baseline on ccnet_perplexity. It confirms the practical consequence of deviating from CCNet’s original per-language design intent (Wenzek et al., 2019) by precisely quantifying the resulting disparity. It provides the V baseline that Turki et al. [2026]’s endorsement of percentile-based per-language thresholding implicitly requires for rigorous evaluation.

6.0.2. 6.2 LIMITATIONS

L1: Existence-only scope. This paper tests only the existence and magnitude of the disparity under global thresholding. Whether per-language calibration reduces the disparity substantially ($\Delta V \geq 0.10$ for $\geq 3/5$ k values) remains to be tested in follow-on work. The characterization contribution is complete and independently publishable as a necessary baseline.

L2: Sample scope. Results are based on 208,262 documents, approximately 0.0002% of the full 113.3 billion document RedPajama-V2 corpus. The phenomenon is structurally driven and confirmed across three independent runs on the same sample; extrapolation to the full corpus requires distributed-scale replication.

L3: Document-length confound. Longer documents tend to have lower perplexity. Length stratification was planned but not implemented in the current analysis. The 72.7 percentage-point max–min retention gap substantially exceeds what a plausible length confound alone can explain; however, length stratification is a recommended robustness check for follow-on work.

L4: Five European languages. The five-language

sample covers only the Germanic (en, de) and Romance (es, fr, it) families within Indo-European. Findings may not generalize to low-resource, agglutinative, or non-Indo-European languages, which may exhibit different or more severe distribution divergences from Wikipedia-trained KenLM models.

6.0.3. 6.3 BROADER IMPACT

Any model trained on a corpus filtered using global percentile thresholds on ccnet_perplexity will have structurally imbalanced language coverage, favoring Romance-family languages over Germanic-family languages regardless of the specific threshold chosen. This work enables practitioners to: (a) audit existing multilingual training datasets for this bias source, (b) design properly powered correction studies using the $V = 0.40\text{--}0.57$ calibrated baseline, and (c) adopt per-language calibration as a default practice. Correction strategies should be evaluated against the baseline established here before deployment in production pipelines.

7. Conclusion

A practitioner applying CCNet’s standard perplexity filter to RedPajama-V2 retains 86.4% of Spanish documents but only 16.3% of English documents at a moderate threshold — not from any quality difference between the two language groups, but from a calibration artifact introduced by applying a global threshold to perplexity scores that were produced by per-language models trained on incommensurable reference corpora. This paper characterizes that artifact precisely: a large, structural, language-family retention divide (Cramér’s $V = 0.40\text{--}0.57$) that is consistent across the full practical range of filtering aggressiveness, maps exactly onto the Romance–Germanic divide, and is 25–40% larger than prior estimates suggested.

Three contributions are reported: (1) the first calibrated Cramér’s V baseline for global ccnet_perplexity thresholding on RedPajama-V2, (2) quantification of the language-family structure of the disparity consistent with the CCNet KenLM mechanism, and (3) recalibration of prior effect size estimates that affects how researchers should design correction studies.

The most urgent next step is testing whether per-language k -th percentile calibration reduces $\Delta V \geq 0.10$ for ≥ 3 of 5 k values — the direct correction hypothesis. If confirmed, per-language calibration should become the default rather than the exception in multi-

lingual dataset curation. The challenge of calibrating multilingual quality filters equitably has been recognized qualitatively for years; this work provides the quantitative foundation needed to evaluate progress against a precise, reproducible standard.

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